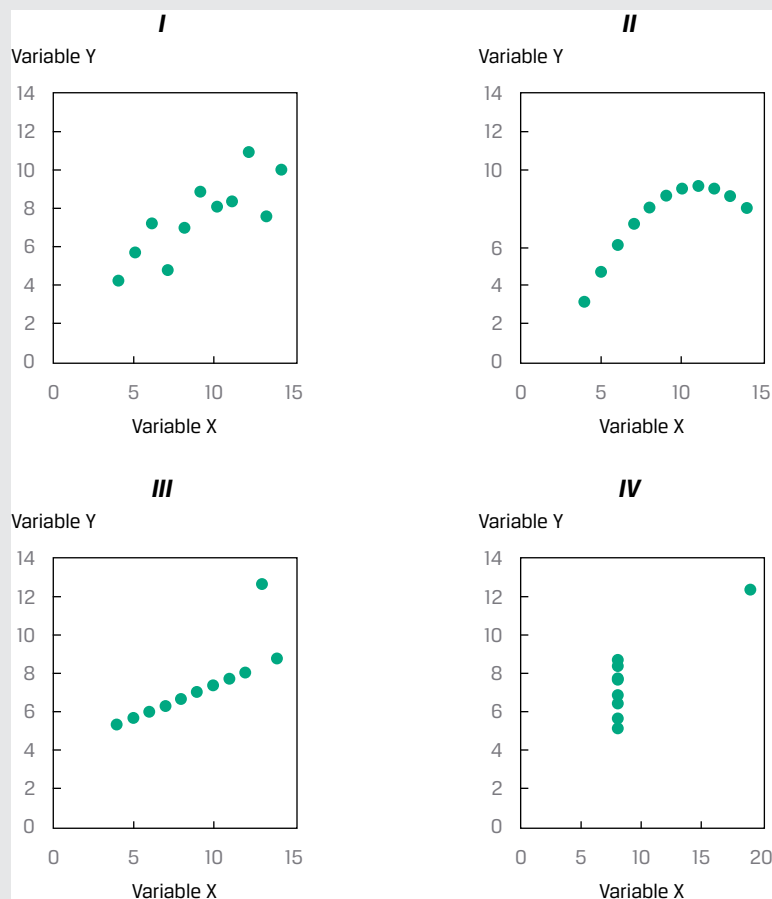


Exhibit 28: Visual Depiction

The bottom line? Knowing the means and standard deviations of the two variables, as well as the correlation between them, does not tell the entire story.

Source: Francis John Anscombe, "Graphs in Statistical Analysis," *American Statistician* 27 (February 1973): 17–21.

QUESTION SET

1. A correlation of 0.34 between two variables, X and Y , is *best* described as:

- A. changes in X causing changes in Y .
- B. a positive association between X and Y .
- C. a curvilinear relationship between X and Y .

Solution:

B is correct. The correlation coefficient is positive, indicating that the two series move together.

2. Which of the following is a potential problem with interpreting a correlation coefficient?

- A. Outliers
- B. Spurious correlation